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Call Center Analytics Report

Understanding and Predicting Call Handling Time

Prepared for: Board of Directors
Data source: 50,000 call center records
August 2026

60% PREDICTION ACCURACY <i>of call-length variation explained</i>	~1.3 min TYPICAL ERROR <i>vs. 3.1 min guessing the average</i>	50,000 CALLS ANALYZED <i>across 6 months of operations</i>
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Executive Summary

This report analyzes 50,000 customer call records to understand what drives how long a call lasts, and whether that duration can be predicted in advance. The goal is to give leadership a data-backed view of where time is being spent on calls, and where operational changes could shorten handling time.

The average call lasts about 8.4 minutes. Using a statistical model built from the data, we can predict a given call's length to within roughly 1.3 minutes on average, more than twice as accurate as simply assuming every call takes the average time. The model also shows clearly which factors add the most time to a call, and which do not.

Three headline findings:

- Issue type is by far the biggest factor. Network, roaming and billing issues take considerably longer to resolve than the average call, in some cases adding 5 to 7 extra minutes.
- Agent experience shortens calls. Each additional year of agent experience is associated with a measurable reduction in call time, pointing to training and retention as a lever for efficiency.
- Repeat contact and chatbot handoffs add time. Customers who called before within the last 30 days, and calls that were transferred from the chatbot to a human agent, both run longer, suggesting friction in first-contact resolution and handoff.

Business Context & Objective

Call duration directly drives staffing needs, wait times, and cost to serve. Understanding what makes calls longer or shorter helps the business plan staffing more accurately, identify processes that are consuming excess agent time, and set realistic service expectations for customers.

This analysis used 50,000 call records covering issue type, customer segment, subscription type, region, agent experience, prior contact history, and operational conditions (such as network incidents, M-PESA outages, and holidays). A statistical model was built to predict call duration from these factors, and to quantify each factor's individual effect.

What the Data Shows

Call Duration Today

The typical call lasts 8.4 minutes. Most calls fall between 5 and 15 minutes, though a smaller number of complex calls run considerably longer, pulling the average up.

Factors Most Related to Call Length

Before building a predictive model, we looked at which factors move together with call duration most closely:

- Repeat callers (customers who contacted support multiple times in the past 30 days) tend to have longer calls, the strongest single relationship in the data.
- Longer queue waits are associated with modestly longer calls.
- More experienced agents are associated with shorter calls, the strongest factor linked to reducing call time.

No factor in the dataset was found to overlap or duplicate another (checked statistically via variance inflation), meaning each effect described in this report can be interpreted on its own rather than being distorted by a hidden, related factor.

How Well Can We Predict Call Duration?

We tested two modelling approaches: a linear regression model (which estimates a straight-line effect for each factor) and a random forest model (which can capture more complex, non-linear patterns). Both were evaluated on a held-out set of calls the model had not seen during training, and compared against a naive baseline of simply guessing the overall average every time.

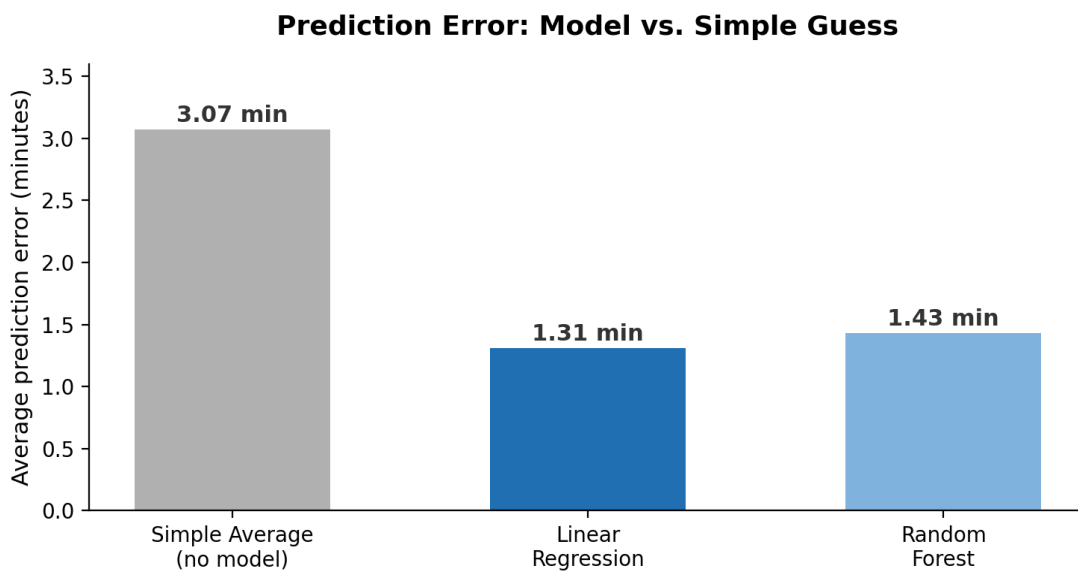


Figure 1: Average prediction error, in minutes. Lower is better.

The linear regression model performed best, cutting the average prediction error from 3.07 minutes (naive guess) down to 1.31 minutes, a 57% reduction. The more complex random forest model did not improve on this, suggesting the relationships in the data are largely straightforward and additive rather than involving hidden, complex interactions. This is a favorable result: the simpler model is also the easier one to explain and act on.

Overall, the model explains about 60% of the variation in call duration ($R^2 = 0.60$). The remaining 40% reflects factors not captured in this dataset, such as the specific complexity of an individual issue, the conversation itself, or agent-to-agent variation not tied to tenure.

Model Performance at a Glance

Metric	Naive Guess	Linear Regression	Random Forest
Typical error (MAE)	3.07 min	1.31 min	1.43 min
Larger-miss error (RMSE)	4.22 min	2.66 min	2.85 min
Variation explained (R^2)	0%	60%	54%

MAE = mean absolute error (typical size of a miss). RMSE = root mean squared error (penalizes large misses more heavily). Both are in minutes.

What Drives Call Duration

The chart below shows the estimated effect of each factor on call length, holding other factors constant. Bars to the right add time to a call; the bar to the left (agent experience) reduces it.

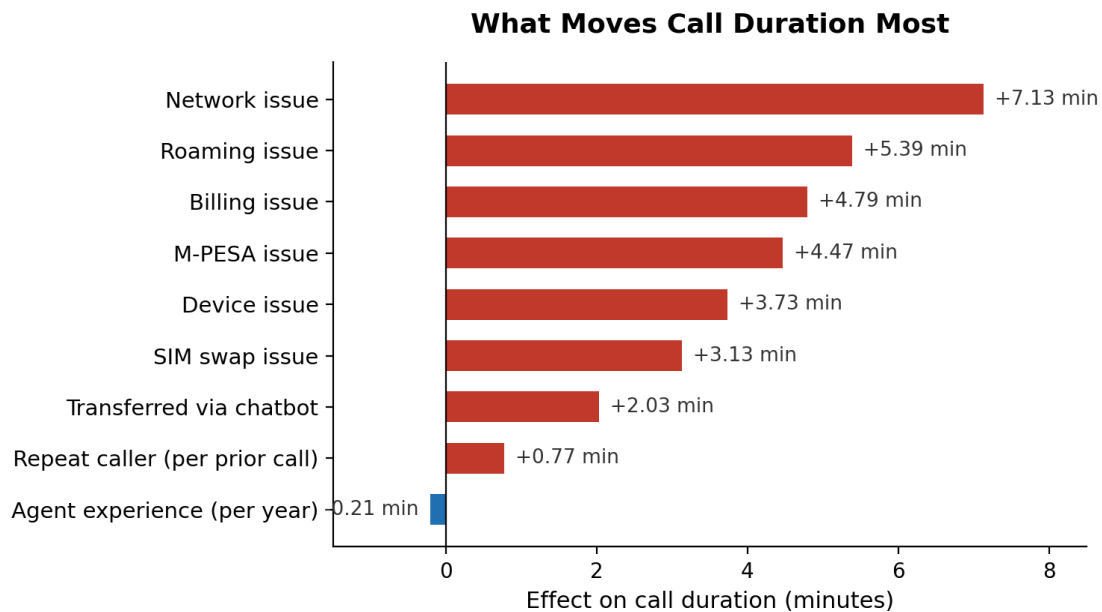


Figure 2: Estimated minutes added (or saved) per unit of each factor.

Practical reading of these numbers:

- A network-related issue takes roughly 7 minutes longer than the baseline issue type, the single largest driver of handling time.
- Roaming, billing, M-PESA, device, and SIM-swap issues each add between 3 and 5.5 minutes.

- A call that gets transferred from the chatbot to a human agent runs about 2 minutes longer than one that reaches an agent directly.
- Each prior call from the same customer in the last 30 days adds roughly 45 seconds, repeat contact compounds handling time.
- Each additional year of agent experience shaves about 12 seconds off the average call, small per call, but meaningful at scale across thousands of calls per month.

Recommendations

- Prioritize agent training and retention. Experience reliably shortens calls; investing in ramp-up time and retaining tenured agents compounds into real handling-time savings.
- Review network, roaming, and billing support workflows. These issue types drive the largest amount of extra handling time and are strong candidates for dedicated scripts, tools, or specialist queues.
- Audit the chatbot-to-agent handoff. Transferred calls taking longer may indicate the chatbot is not capturing enough context before escalating, causing agents to re-ask questions.
- Target repeat callers for first-contact resolution. Customers calling multiple times in 30 days are a signal of unresolved issues; proactive follow-up could reduce overall call volume and time.
- Use the model for staffing forecasts, with the caveat that it explains 60% of variation. It is reliable for planning at the aggregate level, but should not be used to judge individual agents or calls in isolation.

Notes & Limitations

- This analysis identifies patterns and associations in historical data, not proof of cause and effect. For example, experienced agents may also tend to handle different types of calls.
- The model explains 60% of why calls vary in length; the remaining 40% reflects factors not captured in this dataset, such as the specific nature of the customer's issue or the conversation itself.
- Results reflect the 50,000 calls analyzed and may shift as customer behavior, products, or support processes change over time. Periodic refresh of the model is recommended.